

Lecture 9: Risk Aversion vs Intertemporal Substitution

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February, 2026

Intro

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Building Blocks

What is Risk Aversion?

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- You don't need the concept of *time* to think about risk aversion

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- You don't need the concept of *time* to think about risk aversion
- X : a set of outcomes; $\Delta(X)$: the set of lotteries over X (the simplex)
- Let $\succeq$ be a preference relation over $\Delta(X)$
- You are **risk-averse** if, for any $p \in \Delta(X)$, you have $\delta_{\mathbb{E}[p]} \succeq p$

What Von-Neumann and Morgenstern Taught Us

- Typical assumptions: $\succeq$ is complete, transitive, and continuous
- Independence axiom: if $p \succeq q$, then for any r and $\alpha \in (0, 1)$, we have $\alpha p + (1 - \alpha)r \succeq \alpha q + (1 - \alpha)r$

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- Under independence, $\succeq$ can be represented by a utility function $U : \Delta(X) \rightarrow \mathbb{R}$ such that

$$U(p) = \sum_{x \in X} \pi(x)u(x)$$

where $\pi(x)$ is the probability of outcome x and $u(x)$ is the utility of outcome x

- The preferences we will cover in this lecture will violate exactly this axiom, stay tuned!

More or Less Risk Averse?

- The **Certainty Equivalent** of $p \in \Delta(X)$ is the amount of money c such that $\delta_c \sim p$

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- John is (weakly) **more risk-averse** than Mary if $CE_J(p) \leq CE_M(p)$ for all $p \in \Delta(X)$
- Pratt showed that this is all about how concave u is!
- Concavity is measured by second derivatives... but these are not scale-free!
- We typically measure risk aversion by $-u''(\cdot)/u'(\cdot)$ or $-x \cdot u''(x)/u'(x)$
- Many prominent utility functions: CARA, CRRA, HARA, ...

The Special Case of CRRA

- In Macro, we often use the CRRA utility function:

$$u(C) = \frac{C^{1-\gamma} - 1}{1-\gamma}$$

- The parameter γ is the coefficient of relative risk aversion (RRA)

$$\frac{-c \cdot u''(c)}{u'(c)} = \gamma$$

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- If $x \succeq y$, then $\alpha x \succeq \alpha y$ for any $\alpha > 0$ (Indifference curves are radially symmetric)
- Optimal portfolio weights are independent of wealth (with complete markets)
- Equilibrium prices (and returns) depend on *growth rates*, not levels

Questions?

Intertemporal Substitution

- You don't need the concept of *uncertainty* to think about intertemporal substitution
- Think about two periods and a consumer choosing optimal consumption:

$$\max_{C_1, C_2} u(C_1) + \beta u(C_2) \quad \text{s.t.} \quad C_1 + \frac{C_2}{1+r} = W$$

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- **Elasticity of Intertemporal Substitution (EIS):** $\frac{d \log(c_2/c_1)}{d \log(1+r)} > 0$

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- **Elasticity of Intertemporal Substitution (EIS):** $\frac{d \log(c_2/c_1)}{d \log(1+r)} > 0$
- Exercise: show that for CRRA preferences, the EIS is $1/\gamma$

Risk Aversion vs Intertemporal Substitution

- With CRRA preferences, high $\gamma \implies$ high risk aversion
- But high $\gamma \implies$ low EIS $\implies$ low willingness to substitute consumption
- Only two possibilities:
 - High γ : very risk-averse, hates shifting consumption around
 - Low γ : not very risk-averse, and very willing to shift consumption around

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- Only two possibilities:
 - High γ : very risk-averse, hates shifting consumption around
 - Low γ : not very risk-averse, and very willing to shift consumption around
- If you bump up γ to make the consumer afraid of equity...
- She will really upset about shifting consumption around!
- You have to make the relative price of consuming tomorrow low $\implies$ high risk-free rate!
- The prize for waiting must be large... risk-free rate puzzle from Weil (1989)!

Questions?

A New Class of Preferences Comes to Town

Do We Care About *When* Uncertainty is Resolved?

- With expected utility, only probabilities and outcomes matter
- The timing of uncertainty is irrelevant
- But what if you care about *when* uncertainty is resolved?

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- But what if you care about *when* uncertainty is resolved?
- An irreversible medical condition... do you want to find out now or later?
- At the slot machine in Vegas... do you want to wait and gamble, or to resolve it immediately?
- It's already decided whether you get tenured or not... do you want to find out now or later?

Kreps-Porteus Preferences

- Kreps and Porteus (1978): axiomatic characterization of preferences over lotteries *and* timing
- In the EU framework, “time” only matters because it induces other *states*...
- Example: flip a coin, consume 5 at $t = 0$, 10 at $t = 1$ if heads, 0 at $t = 1$ if tails
- Do you want to find out the outcome of the coin flip at $t = 0$ or at $t = 1$?

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- Under EU, you should be indifferent between options
- KP preferences allow you to prefer one over the other
- EU: preferences over probability distributions
- KP: preferences over “filtrations” – information structures
- KP recover EU when the agent does not care about timing

Quick Look Into Their Setup (Skipping Super Technical Terms)

- Time is discrete $t \in \{0, 1, 2, \dots, T\}$
- For every t , there is a payoff space Z_t
- D_T is the set of probability measures over Z_T (lotteries at time T)

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- X_t : the set of all nonempty closed subsets of D_t
- $D_{T-1} \equiv Z_{T-1} \times X_T$

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- $D_{T-1} \equiv Z_{T-1} \times X_T$
- In general: $D_t \equiv Z_t \times X_{t+1}, \forall t < T$
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- $D_{T-1} \equiv Z_{T-1} \times X_T$
- In general: $D_t \equiv Z_t \times X_{t+1}, \forall t < T$
- Your preference relation $\succeq$ ranks elements of D_t for all t
- Usual stuff: completeness, transitivity, continuity, monotonicity, ...
- If you want some stuff, check the paper!

Example

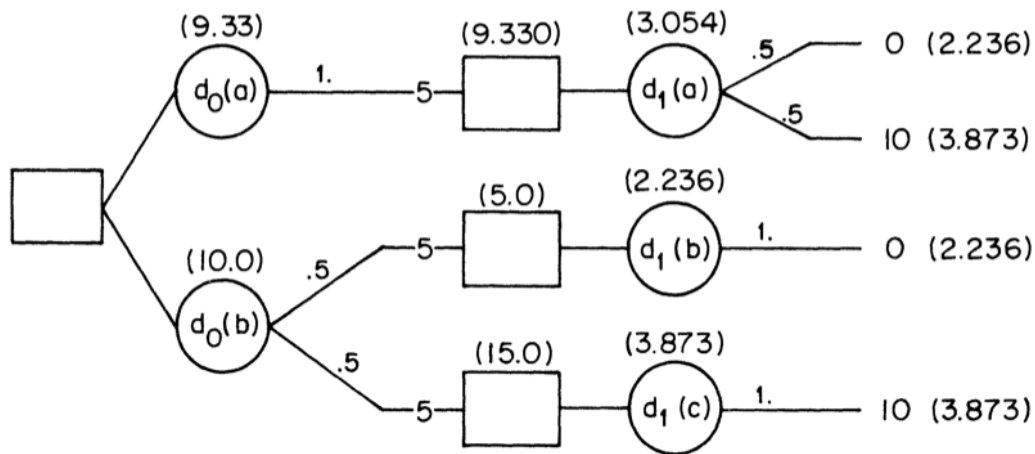


FIGURE 2.

The Representation Theorem

- In a finite decision tree, suppose U_t is your continuation value
- U_{t+1} : the *random* (from the point of view of t) continuation value tomorrow
- Let $\mu_t(U_{t+1})$ be the certainty equivalent of U_{t+1}
- Interpretation: money you get right now, for sure, to stop playing

The Representation Theorem

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- Interpretation: money you get right now, for sure, to stop playing
- Axioms from Kreps and Porteus (1978) $\implies$ there exists an “aggregator” function f such that

$$U_t = f(Z_t, \mu_t(U_{t+1}))$$

- $\mu_t(\cdot)$ encodes how risk-averse you are: the more risk-averse, the lower $\mu_t(U_{t+1})$ is
- The aggregator measures how much you trade-off payoff today vs keep playing
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- If f is linear in the second argument and $\mu_t(U_{t+1}) = \mathbb{E}_t[U_{t+1}]$ we recover expected utility!

Epstein and Zin (1989)

- L. Epstein and Zin ([1989](#)) made the whole setup operational
- They extended Kreps and Porteus ([1978](#)) to the infinite horizon case
- Super non-trivial since there is no “end date”... continuation becomes a fixed point problem

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- Super non-trivial since there is no “end date”... continuation becomes a fixed point problem
- They specialize functional forms:

$$f(x, y) \equiv ((1 - \beta)x^{1-\rho} + \beta y^{1-\rho})^{\frac{1}{1-\rho}}$$
$$\mu_t(U_{t+1}) \equiv (\mathbb{E}_t[U_{t+1}^{1-\alpha}])^{\frac{1}{1-\alpha}}$$

- α controls risk aversion, ρ controls intertemporal substitution

Heuristic Derivation of the SDF

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- L. G. Epstein and Zin ([1991](#)) took it to the data and showed how to use the implied SDF
- We will follow a heuristic derivation from François Gourio's lecture notes (see Github)
- We will assume, just to make math clear, that we have a countable state space
- Expectations will become sums weighted by conditional probabilities $\pi_t(s_{t+1})$

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$$\mu_t(U_{t+1}) = \left(\sum_{s_{t+1}} \pi_t(s_{t+1}) U_{t+1}(s_{t+1})^{1-\alpha} \right)^{\frac{1}{1-\alpha}}$$

The Recursion

- Substituting μ_t into f , the Epstein-Zin recursion is:

$$U_t = \left[(1 - \beta)C_t^{1-\rho} + \beta \left(\sum_{s_{t+1}} \pi_t(s_{t+1}) U_{t+1}(s_{t+1})^{1-\alpha} \right)^{\frac{1-\rho}{1-\alpha}} \right]^{\frac{1}{1-\rho}}$$

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- Let $q_t(s_{t+1})$ be the time- t price of an Arrow-Debreu security
- If you buy one unit, your utility changes by $\partial U_t / \partial C_{t+1}(s_{t+1})$
- You pay $q_t(s_{t+1})$ and each unit of consumption hurts you by $\partial U_t / \partial C_t(s_t)$

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- At the margin $\frac{\partial U_t}{\partial C_{t+1}(s_{t+1})} = q_t(s_{t+1}) \cdot \frac{\partial U_t}{\partial C_t(s_t)}, \forall s_{t+1}, s_t, t$
- For any asset i with return $R_{i,t+1}(s_{t+1})$ in state s_{t+1} , we have

$$\mathbb{E}_t [M_{t+1} R_{i,t+1}] = \sum_{s_{t+1}} \pi_t(s_{t+1}) M_{t+1}(s_{t+1}) R_{i,t+1}(s_{t+1}) = \sum_{s_{t+1}} q_t(s_{t+1}) \cdot R_{i,t+1}(s_{t+1}) = 1$$

Marginal Utilities

- Raise both sides of the recursion to the power $1 - \rho$:

$$U_t^{1-\rho} = (1 - \beta) C_t^{1-\rho} + \beta \mu_t^{1-\rho}$$

- Differentiate w.r.t. C_t :

$$\frac{\partial U_t}{\partial C_t} = (1 - \beta) \left(\frac{C_t}{U_t} \right)^{-\rho}$$

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- For $C_{t+1}(s_{t+1})$, chain rule:
$$\frac{\partial U_t}{\partial C_{t+1}(s_{t+1})} = \frac{\partial U_t}{\partial \mu_t} \cdot \frac{\partial \mu_t}{\partial U_{t+1}(s_{t+1})} \cdot \frac{\partial U_{t+1}(s_{t+1})}{\partial C_{t+1}(s_{t+1})}$$

$$\frac{\partial U_t}{\partial \mu_t} = \beta \left(\frac{\mu_t}{U_t} \right)^{-\rho}, \quad \frac{\partial \mu_t}{\partial U_{t+1}(s_{t+1})} = \pi_t(s_{t+1}) \left(\frac{U_{t+1}(s_{t+1})}{\mu_t} \right)^{-\alpha},$$
$$\frac{\partial U_{t+1}(s_{t+1})}{\partial C_{t+1}(s_{t+1})} = (1 - \beta) \left(\frac{C_{t+1}(s_{t+1})}{U_{t+1}(s_{t+1})} \right)^{-\rho}$$

The ratio of marginal utilities yields:

$$q_t(s_{t+1}) = \frac{\partial U_t / \partial C_{t+1}(s_{t+1})}{\partial U_t / \partial C_t(s_t)} = \beta \pi_t(s_{t+1}) \left(\frac{C_{t+1}(s_{t+1})}{C_t} \right)^{-\rho} \left(\frac{U_{t+1}(s_{t+1})}{\mu_t} \right)^{\rho-\alpha}$$

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- When $\alpha = \rho$: the last term vanishes and we recover standard CRRA

$$M_{t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\rho}$$

- Problem: $U_{t+1}(s_{t+1})/\mu_t$ is not directly observable
- We will deal with this later using the return on the “wealth portfolio”

Questions?

The Wealth Portfolio Trick

- Recall: we derived

$$M_{t+1}(s_{t+1}) = \beta \left(\frac{C_{t+1}(s_{t+1})}{C_t} \right)^{-\rho} \left(\frac{U_{t+1}(s_{t+1})}{\mu_t} \right)^{\rho-\alpha}$$

- The ratio $U_{t+1}(s_{t+1})/\mu_t$ involves the continuation value — not observable!
- Idea:** link it to the return on a traded portfolio

- Let W_t be total wealth: the value of all future consumption
- Budget constraint (Gourio, Section 3):

$$W_{t+1}(s_{t+1}) = (W_t - C_t) R_{a,t+1}(s_{t+1})$$

- Consume C_t out of W_t , invest the rest at the return $R_{a,t+1}$

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- Consume C_t out of W_t , invest the rest at the return $R_{a,t+1}$
- By homotheticity of EZ preferences: U_t is homogeneous of degree 1 in W_t
- Value function takes the form $U_t = \phi(s_t) W_t$, i.e., ϕ depends on the state, not on wealth

- Using $U_{t+1}(s_{t+1}) = \phi(s_{t+1}) (W_t - C_t) R_{a,t+1}(s_{t+1})$:

$$\frac{U_{t+1}(s_{t+1})}{\mu_t} = \frac{\phi(s_{t+1}) R_{a,t+1}(s_{t+1})}{\left(\mathbb{E}_t \left[\phi(s_{t+1})^{1-\alpha} R_{a,t+1}(s_{t+1})^{1-\alpha} \right] \right)^{\frac{1}{1-\alpha}}}$$

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- Note: $(W_t - C_t)$ is known at t and cancels immediately
- But $\phi(s_{t+1})$ is **random** — it does not cancel!
- We need one more step...

- Define $\Phi_t \equiv (\mathbb{E}_t [(\phi(s_{t+1}) R_{a,t+1})^{1-\alpha}])^{1/(1-\alpha)}$, so $\mu_t = (W_t - C_t) \Phi_t$
- FOC for C_t (optimality of consumption-savings):

$$(1 - \beta) C_t^{-\rho} = \beta (W_t - C_t)^{-\rho} \Phi_t^{1-\rho}$$

FOC and Envelope Theorem

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- Envelope theorem + FOC $\implies \phi(s_t) = \partial U_t / \partial C_t = (1 - \beta)(C_t / U_t)^{-\rho}$
- With $U_t = \phi(s_t) W_t$:

$$\phi(s_t)^{1-\rho} = (1 - \beta) \left(\frac{W_t}{C_t} \right)^{\rho}$$

The Key Identity

- Envelope at $t + 1$: $\phi(s_{t+1})^{1-\rho} = (1 - \beta)(W_{t+1}/C_{t+1})^\rho$
- FOC rearranged: $\Phi_t^{1-\rho} = \frac{1-\beta}{\beta} \left(\frac{W_t - C_t}{C_t} \right)^\rho$
- Take the ratio and use $R_{a,t+1} = W_{t+1}/(W_t - C_t)$:

$$\frac{\phi(s_{t+1})^{1-\rho}}{\Phi_t^{1-\rho}} = \beta R_{a,t+1}^\rho \left(\frac{C_t}{C_{t+1}} \right)^\rho$$

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- Take the ratio and use $R_{a,t+1} = W_{t+1}/(W_t - C_t)$:

$$\frac{\phi(s_{t+1})^{1-\rho}}{\Phi_t^{1-\rho}} = \beta R_{a,t+1}^\rho \left(\frac{C_t}{C_{t+1}} \right)^\rho$$

- Since $\left(\frac{U_{t+1}}{\mu_t} \right)^{1-\rho} = \frac{\phi(s_{t+1})^{1-\rho}}{\Phi_t^{1-\rho}} R_{a,t+1}^{1-\rho}$:

$$\boxed{\left(\frac{U_{t+1}}{\mu_t} \right)^{1-\rho} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\rho} R_{a,t+1}}$$

- All ϕ terms have cancelled!

The Epstein-Zin SDF

- Define $\theta \equiv \frac{1-\alpha}{1-\rho}$, and note $\rho - \alpha = (1-\rho)(\theta - 1)$
- Raise the key identity to power $\theta - 1$:

$$\left(\frac{U_{t+1}}{\mu_t} \right)^{\rho-\alpha} = \beta^{\theta-1} \left(\frac{C_{t+1}}{C_t} \right)^{-\rho(\theta-1)} R_{a,t+1}^{\theta-1}$$

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- $\alpha = \rho$ ($\theta = 1$): $M_{t+1} = \beta \left(\frac{C_{t+1}}{C_t}\right)^{-\rho}$ — standard CRRA
- $\alpha \neq \rho$: SDF depends on consumption growth **and** the wealth return

Meeting the Data

Can Kreps-Porteus Preferences Solve the Equity Premium Puzzle?

- Weil (1989): no, not really. You need way more than only that

Net risk premium and net risk-free rate ^a ($\beta = 0.98$)							
EIS ($1/\rho$)	CRRA (γ)						
	0	0.5	1	5	10	20	45
∞	0.00	0.05	0.09	0.47	0.93	1.76	3.00
	2.04	2.02	1.99	1.80	1.58	1.21	0.91
2	0.01	0.06	0.11	0.51	1.00	1.88	3.13
	2.95	2.91	2.87	2.55	2.17	1.53	0.75
1	0.01	0.07	0.12	0.55	1.07	2.00	3.27
	3.86	3.81	3.75	3.31	2.77	1.85	0.59
0.2	0.10	0.18	0.26	0.89	1.65	2.93	4.37
	11.49	11.29	11.10	9.56	7.72	4.45	-0.68
0.1	0.25	0.36	0.47	1.33	2.37	4.10	5.79
	21.87	21.47	21.08	17.95	14.26	7.83	-2.24
0.05	0.59	0.75	0.89	2.18	3.73	6.37	8.82
	45.89	44.95	43.98	36.92	28.72	15.01	-5.22
1/45	1.19	1.42	1.66	3.70	6.40	11.50	17.53
	131.52	128.87	125.06	101.02	75.11	36.06	-12.00

^aIn percent The bold number is the net risk-free rate

- There are two more fundamental reasons for the puzzle:
 - There is not enough aggregate risk in consumption growth
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- There are two more fundamental reasons for the puzzle:
 - There is not enough aggregate risk in consumption growth
 - The average consumption growth is too high to be consistent with low interest rates
- Increasing risk in consumption growth: disasters, long-run risks, ...
- Or introduce very imperfect insurance markets (e.g. incomplete markets, borrowing constraints, ...)
- Heterogeneity across agents is likely important!

- Campbell, Chap. 6

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